

# Executive Summary

Predicting Customer Booking Completion

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## 1. Executive Overview

- **Purpose:** Developed a machine learning model (Random Forest) to predict whether a customer will complete a flight booking, enabling proactive marketing.
- **Dataset Scope:** Analyzed 50,000 customer sessions across 14 operational and customer behavior attributes.
- **Conversion Baseline:** ~15.0% conversion rate across historical sessions.

## 2. Model Performance & Accuracy

- **Predictive Power:** Achieved a Cross-Validated ROC-AUC score of 0.761 (0.768 on Test).
- **Key Metric:** Captured 62.8% of all completed bookings (Recall), providing a strong foundation for targeted customer acquisition.

## 3. Top Predictive Drivers

- **Booking Origin & Route Volume (42.4% Importance):** Geographic market and route frequency are the primary determinants of booking intent.
- **Lead Time & Trip Length (27.9% Importance):** Customers booking further in advance or for longer trips show significantly higher purchase commitment.
- **Flight Duration (8.1% Importance):** Haul type directly impacts buying cycles.

## 4. Strategic Recommendations for British Airways

- **Proactive Retargeting:** Trigger personalized email/app incentives for high-intent routes and long-lead customers before they leave the booking channel.
- **Market-Specific Campaigns:** Tailor acquisition campaigns based on high-converting origin countries (e.g., origin frequency signals).